Workbook AZ-900 Bootcamp Microsoft Azure Fundamentals

Cloud Computing

An Intro to Cloud Computing

Cloud computing represents a significant shift in the way businesses approach IT resources. It involves the delivery of various computing services such as servers, storage, databases, and networking over the Internet. This approach allows for faster innovation, flexible resources, and economies of scale. Students will learn that they typically pay only for the cloud services they use, which can lead to reduced operating costs and more efficient infrastructure management.

The lecture emphasizes several key aspects of cloud computing:

- **Cost**: By eliminating the capital expense of buying hardware and software and managing on-site data centers, cloud computing can be a more cost-effective solution.
- **Speed**: Cloud services are often self-service and on-demand, allowing vast amounts of computing resources to be provisioned quickly.
- **Global Scale**: The ability to scale elastically is a significant benefit, meaning that the right amount of IT resources can be delivered when needed, from the right geographic location.
- **Productivity**: Cloud computing reduces the need for many time-consuming IT management tasks, allowing IT teams to focus on more important business goals.
- **Performance**: Utilizing a worldwide network of secure data centers that are regularly upgraded ensures fast and efficient computing hardware, reducing network latency and providing greater economies of scale.
- **Reliability**: Cloud computing enhances data backup, disaster recovery, and business continuity by mirroring data at multiple redundant sites on the cloud provider's network.
- **Security**: Many cloud providers offer robust security measures that help protect data, applications, and infrastructure from potential threats.

The lecture also provides an exam tip for students, emphasizing the importance of understanding the delivery of computing services over the Internet and remembering the key benefits reviewed in the lesson.

The Shared Responsibility Model

In the lecture, the shared responsibility model in the context of cloud computing is explored. This model delineates the security tasks and responsibilities that are handled by the cloud provider and those that are handled by the user. The lecture emphasizes that the workload responsibilities vary depending on the type of service, such as Software as a Service (SaaS), Platform as a Service (PaaS), or Infrastructure as a Service (IaaS). In an on-premises data center, the customer owns the entire stack, but as you move to the cloud, some responsibilities transfer to the provider, such as Microsoft. The lecture also stresses that regardless of the deployment type, ownership of data and identities, and the responsibility for their security, always remain with the user.

Cloud Models and Their Use Cases

The lecture explores the three primary deployment models for cloud computing: public cloud, private cloud, and hybrid cloud. Each model has unique characteristics and considerations that must be taken into account when migrating to the cloud.

The **public cloud** model offers services over the public internet, accessible to anyone who wishes to purchase them. Resources such as servers and storage are owned and operated by a third-party cloud service provider and delivered over the internet. This model does not require capital expenditures to scale up, and applications can be quickly provisioned and deprovisioned. Organizations pay only for what they use.

The **private cloud** model consists of computing resources used exclusively by users from one business or organization. It can be physically located on-site or hosted by a third-party provider. The key takeaway is that private clouds are used by a single organization, and hardware must be purchased for start-up and maintenance. Organizations have complete control over resources and security and are responsible for hardware maintenance and updates.

The **hybrid cloud** model is a computing environment that combines both public and private clouds, allowing data and applications to be shared between them. It is often used by large organizations and provides the most flexibility. Organizations can determine where to run their applications and control security, compliance, or legal requirements.

The Consumption-Based Model

The lecture delves into the consumption-based model in cloud computing, a model where end users pay only for the resources they use. This approach offers several benefits, including no upfront costs, no need to purchase or manage infrastructure, and the ability to pay for additional resources only when needed. It also allows organizations to stop paying for resources that are no longer required. The consumption-based model is succinctly described as paying for what is used, providing flexibility and efficiency in resource management.

The Benefits of Cloud Computing

Cloud computing offers several advantages over traditional physical environments, such as on-premises data centers. One of the key benefits is high availability, ensuring a continuous user experience without downtime, even during unexpected issues. Scalability is another advantage, allowing applications to scale both vertically (by adding RAM or CPUs) and horizontally (by adding instances of resources). This flexibility ensures that applications can adapt to changing demands.

Elasticity in cloud computing enables applications to auto-scale, adjusting resources as needed. When traffic spikes, the application can increase resources, and when it decreases, the resources can be reduced. This adaptability contributes to the agility of cloud computing, where resources can be deployed and configured quickly as requirements change.

Geo-distribution is another benefit, allowing applications and data to be deployed in regional data centers around the world. This ensures optimal performance and access for end-users in their specific regions. Along with geo-distribution, cloud computing provides robust disaster recovery options through cloud-based backup services and data replication.

From an accounting perspective, cloud computing shifts expenses from capital expenditure (CapEx) to operational expenditure (OpEx). CapEx involves upfront costs for physical infrastructure, while OpEx refers to consumption-based spending on services or products. This shift to OpEx allows organizations to deduct expenses in the same year they are spent, without upfront costs, and aligns with the pay-as-yougo model of cloud services.

Cloud Pricing Models

The lecture on cloud pricing models provides an in-depth examination of the three primary Azure pricing models that cater to various business needs and scenarios. These models include Pay as You Go, Reserved Instances, and Spot Pricing.

The **Pay as You Go** model is characterized by its flexibility, allowing users to pay for Azure services based on actual usage, without any upfront commitment. This model is billed typically per minute or per hour and is ideal for development, testing, short-term projects, and businesses that prefer operational over capital expenditure.

Reserved Instances (RIs) allow users to commit to a specific virtual machine type and size for a fixed term, usually 1 or 3 years, in exchange for discounted pricing. This model offers significant cost savings compared to Pay as You Go and is perfect for long-term projects with predictable resource requirements.

Spot Pricing lets users take advantage of unused Azure capacity at a significant discount. Governed by a bidding system, this model is highly cost-effective but comes without guarantees. Spot instances are suitable for batch processing, data analysis, and non-critical development and testing.

In conclusion, the lecture emphasizes that Azure's three pricing models cater to different needs and use cases, offering flexibility, cost savings, and options for cost-sensitive, interruptible tasks.

Cloud Service Types

Infrastructure-as-a-Service (IaaS)

The lecture on Infrastructure as a Service (IaaS) provides a comprehensive overview of one of the fundamental service models in cloud computing. IaaS offers essential compute, storage, and networking resources on demand, on a pay-as-you-go basis. It's one of the four types of cloud services, along with Software as a Service (SaaS), Platform as a Service (PaaS), and serverless.

IaaS allows organizations to bypass the cost and complexity of buying and managing physical servers and data center infrastructure. Each resource is offered as a separate service component, and users only pay for a particular resource for as long as they need it. The cloud computing service provider, such as Azure, manages the infrastructure, while the organization is responsible for purchasing, installing, configuring, and managing its software, including operating systems, middleware, and applications.

The lecture also highlights common IaaS business scenarios, including lift-and-shift migration, test and development, storage, backup and recovery, web apps, and high-performance computing. These scenarios illustrate the versatility and applicability of IaaS in various organizational contexts.

Platform-as-a-Service (PaaS)

The lecture on Platform as a Service (PaaS) explores a complete development and deployment environment in the cloud. PaaS includes not only infrastructure such as servers, storage, and networking but also middleware, development tools, business intelligence (BI) services, database management systems, and more. It's designed to support the entire web application lifecycle, including building, testing, deploying, managing, and updating.

PaaS allows organizations to avoid the expense and complexity of buying and managing software licenses, underlying application infrastructure, middleware, container orchestrators like Kubernetes, and other resources. The cloud service provider typically manages everything except the applications and services developed by the organization. PaaS also provides a framework that developers can build upon to develop or customize cloud-based applications, reducing the amount of coding required.

Common PaaS scenarios include development frameworks, analytics or business intelligence tools, and the provision of a complete environment that enables organizations to deliver everything from simple cloud-based apps to sophisticated, cloud-enabled enterprise applications.

Software-as-a-Service (SaaS)

The lecture on Software as a Service (SaaS) delves into a cloud computing model that allows users to connect to and use cloud-based applications over the Internet. Common examples include email, calendaring, and office tools. SaaS offers a complete software solution that organizations can purchase on a pay-as-you-go basis from a cloud service provider. The underlying infrastructure, middleware, application software, and data are all managed by the service provider, and users can access the application via a web browser.

SaaS is particularly beneficial for organizations looking to get quickly up and running with an application at minimal upfront cost. It covers everything from free services for personal use to organizational tools such as customer relationship management (CRM), enterprise resource planning (ERP), and document management. The payment structure can be based on subscription or the level of use, and the service provider ensures the availability and security of the application and data.

Describe the Core Architectural Components of Azure

Understanding Azure Regions and Regional Pairs

The lecture focuses on Azure regions and regional pairs, essential components of Microsoft's Azure cloud platform. Regions are vital as they provide flexibility to bring applications closer to users, ensuring better scalability, redundancy, and data residency. Azure has specialized regions for compliance or legal purposes, such as those for U.S. government agencies and specific regions in China.

Availability zones, consisting of one or more datacenters, are physically separate within an Azure region. They are equipped with independent power, cooling, and networking, ensuring that if one zone goes down, the others continue working. Azure also creates region pairs, where each region is paired with another within the same geography, at least 300 miles away. This pairing allows for replication of resources and reduces the likelihood of interruptions due to natural disasters or other significant events.

Region pairs have additional advantages, such as prioritizing one region for quicker restoration during an extensive Azure outage, rolling out planned updates one region at a time, and ensuring data continues to reside within the same geography for legal purposes.

Availability Zones in Azure

The lecture delves into the concept of availability zones within Microsoft's Azure cloud platform. Availability zones are physically separate datacenters within an Azure region, each equipped with independent power, cooling, and networking. This separation ensures that if one zone experiences failure, the others continue working, providing a high level of redundancy and resilience.

Availability zones are connected through high-speed, private fiber-optic networks and are primarily used for virtual machines (VMs), managed disks, load balancers, and SQL databases. They are designed to protect against datacenter failures within a region, and not every region has support for availability zones. Azure offers different categories of services that support availability zones, including zonal services, where resources are pinned to a specific zone, and zone-redundant services, where the platform replicates automatically across zones.

Azure Resources and Resource Groups

The lecture focuses on Azure resources and resource groups, two fundamental elements of the Azure platform. A resource is defined as a manageable item available through Azure, such as virtual machines, storage accounts, web apps, databases, and virtual networks. A resource group is a container that holds related resources for an Azure solution, and all resources must be in a resource group. Resource groups are logical containers for resources deployed on Azure, and a resource can only be a member of a single resource group. Resource groups exist to help manage and organize Azure resources, and they can't be nested. If a resource group is deleted, all resources contained within it are also deleted. Resource groups are also a scope for applying role-based access control (RBAC) permissions.

Understanding Azure Subscriptions

The lecture on "Understanding Azure subscriptions" delves into the concept of Azure subscriptions, which are essential for authenticated and authorized access to Azure products and services.
Subscriptions in Azure define boundaries around products, services, and resources, and there are two types of subscription boundaries: Billing and Access control. The Billing boundary determines how an
Azure account is billed, allowing for separate billing reports and invoices for different requirements. The Access control boundary applies access-management policies at the subscription level, reflecting different organizational structures. The lecture also covers scenarios where additional subscriptions might be created, such as separating environments for development and testing, isolating data for compliance, or managing and tracking costs. Subscription limits and customization of billing to meet specific needs are also discussed.

Management Groups in Azure

The lecture on management groups in Azure provides an in-depth understanding of how organizations can efficiently manage access, policies, and compliance for multiple subscriptions. Azure management groups offer a level of scope

above subscriptions, organizing them into containers and applying governance conditions. All subscriptions within a management group automatically inherit these conditions. This structure allows for enterprise-grade management at a large scale, regardless of the type of subscriptions. The lecture also explains the hierarchy of management groups and subscriptions, the ability to apply policies (such as limiting regions for VM creation), and the importance of Azure AD tenant trust within a management group. Key facts include the support of 10,000 management groups in a single directory, a hierarchy depth of six levels, and the one-parent rule for each management group and subscription.

Describe Azure Compute Services

An Intro to Azure Compute Types

Azure compute is a service that provides on-demand computing resources for running cloud-based applications, including disks, processors, memory, networking, and operating systems. These resources can be made available quickly, and users pay only for what they use. Azure supports various computing solutions and offers compatibility with different platforms like Linux, Windows Server, SQL Server, Oracle, IBM, and SAP.

Among the prominent services in Azure compute are Azure Virtual Machines, VM Scale Sets, Azure Container Instances, Azure App Service, and Azure Functions. Virtual machines emulate physical computers, providing total control over the operating system and environment, making them suitable for custom software or hosting configurations. Virtual machine scale sets enable the deployment and management of identical VMs, allowing for true autoscale to handle varying demand.

Containers and Kubernetes, including Container Instances and Azure Kubernetes Service, offer lightweight, virtualized application environments that can be quickly scaled. Azure App Service facilitates the rapid development, deployment, and scaling of web, mobile, and API apps, providing a platform as a service (PaaS) offering. Azure Functions focus on the code running the service without concern for the underlying platform or infrastructure, often used for quick responses to events or timers.

Azure Container Instances vs Azure Kubernetes Service

The lecture titled "Azure Container Instances vs Azure Kubernetes Service" provides an insightful comparison between Azure Container Instances (ACI) and Azure Kubernetes Service (AKS). The lecture begins by introducing containers as a preferred way to package, deploy, and manage cloud applications. It then delves into Azure Container Instances, highlighting its benefits as the fastest and simplest way to run a container in Azure without managing virtual machines or adopting a higher-level service. ACI is presented as an ideal solution for isolated containers, including simple applications, task automation, and build jobs. The lecture likely further explores the differences, use cases, and scenarios where one service may be preferred over the other.

Azure Virtual Desktop

Azure Virtual Desktop is a desktop and app virtualization service that runs on the cloud. It supports multi-session Windows 10 or 11, integrates with existing Remote Desktop Services (RDS), and offers unified management across Windows operating systems. Key features include security through Azure AD integration, scalability, flexibility, and cost management through pooled resources. Users can connect from any device, and connections are established securely without opening inbound ports.

Key Azure Management Tools

Azure Management Tools encompass a variety of tools designed to deploy, manage, and monitor Azure resources. The Azure Portal is a public website accessed via a web browser, allowing users to create, manage, and monitor Azure services. While it offers a user-friendly interface, it lacks automation for repetitive tasks.

For automation and command-line management, Azure PowerShell is commonly used, especially on Windows. It adds Azure-specific commands to Windows PowerShell, enabling tasks like creating virtual machines. Azure CLI, a cross-platform command-line program, offers similar functionality for Windows, Linux, and macOS.

Azure Cloud Shell provides a browser-based scripting environment, allowing users to choose between PowerShell and Bash. It requires a storage account to function. The Microsoft Azure Mobile app, available for iOS and Android,

enables users to manage and monitor Azure accounts and resources on the go, though with limited functionality compared to other tools.

Rest APIs are service endpoints used by developers to create, retrieve, update, and delete access to Azure services. They are essential when developing solutions that need to communicate with Azure. Lastly, Azure Advisor is a free service within Azure that offers recommendations on high availability, security, performance, and cost. It analyzes deployed services and provides actionable insights to improve the environment and reduce costs.

Azure Networking Services

Getting Started with Azure Virtual Networks

The lecture provides an in-depth understanding of Azure virtual networks (vNets), which enable Azure resources to communicate with each other, the internet, and on-premises client computers. The lecture covers basic vNet concepts such as address space, subnets, regions, and subscriptions. It also explains key networking capabilities, including isolation, segmentation, internet communications, routing, filtering, and connecting virtual networks. Various mechanisms for communication with on-premises resources are discussed, such as point-to-site VPNs, site-to-site VPNs, and Azure ExpressRoute. The lecture emphasizes the importance of understanding these concepts for Azure networking, including how to enable secure communication between Azure resources and how to route and filter traffic.

vNet Peering

The lecture focuses on virtual network peering in Azure, a feature that enables seamless connection between two or more Virtual Networks. This connection makes the virtual networks appear as one for connectivity purposes, and all traffic between them is routed through Microsoft's private network. There are two types of peering: Virtual network peering (within the same Azure region) and Global virtual network peering (across Azure regions). The benefits of using virtual network peering include lowlatency, high-bandwidth connections, the ability to communicate between different virtual networks, data transfer across various Azure elements, and no downtime during or after peering creation. The lecture emphasizes that network traffic between peered virtual networks is private and doesn't require public Internet, gateways, or encryption.

Azure DNS

The lecture introduces Azure DNS, a hosting service for DNS domains that integrates with Microsoft Azure infrastructure. Unlike domain name purchasing, Azure DNS focuses on record management within Azure, offering features such as reliability, availability, security, ease of use, and customization.

Azure DNS's reliability and availability are ensured through Azure's global network of DNS name servers and anycast networking, which provides quick performance and high availability. Security features include Azure role-based access control (Azure RBAC), activity logs, and resource locking. However, it's noted that Azure DNS does not currently support DNSSEC, although alternatives like HTTPS/TLS can often suffice.

Ease of use is emphasized, with Azure DNS managing DNS records for both Azure and external resources. It's integrated into the Azure portal and shares the same credentials, support, and billing. DNS billing is based on the number of hosted DNS zones and received queries, and management can be done through various tools, including the Azure portal, PowerShell cmdlets, and Azure CLI.

The lecture also highlights customizable virtual networks with private domains, allowing custom domain names in private virtual networks. Alias records are supported, enabling references to specific Azure resources, and the alias record set can update itself if the underlying resource's IP address changes.

Azure DNS Private Resolver

The lecture introduces Azure DNS Private Resolver, a service that enables querying Azure DNS private zones from an on-premises environment and vice versa without the need for deploying VM-based DNS servers. Currently in public preview, this service operates within an Azure Virtual Network.

When an Azure DNS Private Resolver is created inside a virtual network, inbound endpoints are established for DNS queries, and the resolver's outbound endpoint processes these queries based on a configurable DNS forwarding ruleset. The architecture requires Azure ExpressRoute or a VPN for DNS resolution between Azure virtual networks and on-premises networks.

The benefits of Azure DNS Private Resolver include fully managed high availability and zone redundancy, cost reduction compared to traditional IaaS solutions, private access to Private DNS Zones, scalability with high performance per endpoint, and compatibility with DevOps tools like Terraform, ARM, or Bicep. The lecture also lists the regions where Azure DNS Private Resolver is available.

Azure VPN Gateway

The lecture on Azure VPN Gateway delves into the concept of Virtual Private Networks (VPNs), which use encrypted tunnels to connect private networks securely over an untrusted network, typically the internet. VPNs are often used to enable branch offices to share sensitive information or to connect onpremises data centers to Azure.

Azure VPN Gateway, a type of virtual network gateway, facilitates various connectivity options:

- Site-to-site connection for linking on-premises data centers to virtual networks.
- Point-to-site connection for connecting individual devices to virtual networks.
- Network-to-network connection for linking virtual networks to other virtual networks.

Two types of VPNs are discussed:

- **Policy-based VPNs**: These use static IP addresses for encryption and support IKEv1 only. They are used with static routing and are suitable for specific scenarios requiring compatibility with legacy devices.
- **Route-based VPNs**: Preferred for on-premises devices, these support IKEv2 and use dynamic routing protocols like Border Gateway Protocol (BGP). They are more adaptable to topology changes and are suitable for connections between virtual networks, point-to-site connections, multisite connections, and coexistence with Azure ExpressRoute gateway.

The lecture also details different VPN gateway sizes (SKUs), each with varying capabilities such as the number of tunnels, throughput benchmarks, and Border Gateway Protocol support.

Azure ExpressRoute

Azure ExpressRoute is a service that facilitates the extension of on-premises networks into the Microsoft cloud through a private connection, bypassing the public internet. This private connection is established with the assistance of a connectivity provider and offers access to various Microsoft cloud services, including Microsoft Azure, Microsoft 365, and others.

ExpressRoute provides several key benefits:

- **Reliability and Speed**: It offers more reliable, faster, and consistent connections compared to typical internet connections.
- **Global Connectivity**: With the premium add-on, it enables global connectivity to Microsoft services across all regions.
- **Dynamic Routing**: It uses Border Gateway Protocol (BGP) for dynamic routing between networks.
- **Redundancy**: Built-in redundancy ensures higher reliability.
- **Quality of Service (QoS)**: This includes support for Skype for Business.

ExpressRoute supports different connectivity models:

- **CloudExchange Colocation**: Allows both Layer 2 and Layer 3 connections between your infrastructure and the Microsoft cloud.

- **Point-to-Point Ethernet Connection**: Provides Layer 2 and Layer 3 connectivity between onpremises sites and Azure.
- **Any-to-Any Connection**: Integrates wide area network (WAN) with Azure, offering Layer 3 connectivity.
- **Directly from ExpressRoute Sites**: Offers dual 100 Gbps or 10-Gbps connectivity directly into Microsoft's global network.

Security is a significant aspect of ExpressRoute, as data doesn't travel over the public internet, reducing exposure to potential risks. It acts as a private connection between on-premises and Azure infrastructure.

Private Links and Private Endpoints

Azure Private Link is a service that facilitates private and secure access to Azure Platform-as-a-Service (PaaS) offerings such as Azure Storage, Azure Cosmos DB, and Azure SQL Database, as well as customerowned or partner services. This is achieved through a private endpoint within your virtual network, utilizing a private IP address.

The private endpoint serves as a network interface that securely connects your virtual network to a service powered by Azure Private Link. This connection ensures that traffic between your virtual network and the Azure service travels exclusively over the Microsoft backbone network, eliminating the need to expose the service to the public internet.

Key benefits of Azure Private Link include:

- **Private Access to Azure Services**: It enables connection to all services that can be utilized as application components in Azure, both from Azure and on-premises networks.
- **On-Premises and Peered Networks Integration**: It allows access to Azure services from onpremises networks over ExpressRoute private peering, VPN tunnels, and peered virtual networks, providing a secure migration path to Azure.
- **Protection Against Data Leakage**: By mapping a private endpoint to a specific instance of a PaaS resource, it ensures that consumers can only connect to that particular resource, thereby minimizing data leakage risks.

Describe Azure Storage Services

An Intro to Azure Storage Accounts

Azure Storage Accounts are essential components in Azure, providing a unique namespace for Azure Storage data objects like blobs, file shares, queues, tables, and disks. These accounts can be accessed globally over HTTP or HTTPS. There are several types of storage accounts, each with different features, redundancy options, and use cases.

Types of Storage Accounts

1. **Standard General-Purpose v2**
    - **Supported Services**: Blob Storage (including Data Lake Storage), Queue Storage, Table Storage, Azure Files.
    - **Redundancy Options**: LRS, GRS, RA-GRS, ZRS, GZRS, RA-GZRS.
    - **Usage**: Recommended for most scenarios using Azure Storage. Supports various storage types.
2. **Premium Block Blobs**
    - **Supported Services**: Blob Storage (including Data Lake Storage).
    - **Redundancy Options**: LRS, ZRS.
    - **Usage**: Suitable for high transaction rates, smaller objects, or consistently low storage latency.

3. **Premium File Shares**
    - **Supported Services**: Azure Files.
    - **Redundancy Options**: LRS, ZRS.
    - **Usage**: Ideal for enterprise or high-performance scale applications. Supports both SMB and NFS file shares.
4. **Premium Page Blobs**
    - **Supported Services**: Page blobs only.
    - **Redundancy Options**: LRS.
    - **Usage**: Specific to page blobs.

Service Level Agreements (SLAs) are attached to storage accounts, and understanding the different types of storage accounts, their supported services, redundancy options, and common use cases is essential for Azure professionals.

Azure Blobs

Azure Blob storage is a versatile object storage solution provided by Microsoft for the cloud. It's designed to handle unstructured data, such as text, images, audio, video, and binary data. Here's an overview of the key aspects of Azure Blob storage:

Key Features

- **Unstructured Data Handling**: Ideal for storing large amounts of data that doesn't fit a specific model or definition.
- **Accessibility**: Accessible via HTTP/HTTPS from anywhere in the world using various tools like Azure Storage REST API, Azure PowerShell, Azure CLI, or client libraries in different languages.
- **Integration with Azure Data Lake Storage Gen2**: Offers a hierarchical file system with the benefits of Blob storage, including low-cost storage, high availability, strong consistency, and disaster recovery.
- **Storage Tiers**: Different access tiers are available to optimize costs based on the frequency of access:
- **Hot Access Tier**: For frequently accessed data.
- **Cool Access Tier**: For infrequently accessed data, stored for at least 30 days.
- **Archive Access Tier**: For rarely accessed data, stored for at least 180 days, with the lowest storage costs.

Blob Storage Resources

1. **Storage Account**: The overarching account that contains all Azure Storage data objects.
2. **Container**: A specific location within the storage account where blobs are organized.
3. **Blob**: The individual data objects stored within a container.

Use Cases

- Serving images or documents to browsers.
- Storing files for distributed access.
- Streaming multimedia content.
- Logging.
- Backup, restore, disaster recovery, and archiving.

- Data analysis by on-premises or Azure-hosted services.

Azure Files

Azure Files is a service that offers fully managed file shares in the cloud, accessible via the Server Message Block (SMB) or Network File System (NFS) protocols. It provides a versatile solution for various scenarios, from replacing traditional file servers to supporting cloud development and containerization. Here's an overview of Azure Files:

Key Features

- **Protocols Supported**: SMB and NFS, making it accessible from Windows, Linux, and macOS clients.
- **Concurrent Mounting**: File shares can be mounted concurrently by cloud or on-premises deployments.
- **Integration with Azure File Sync**: Allows caching on Windows Servers for faster access.
- **Azure Files AD Authentication**: Works with AD hosted on-premises for access control.

Use Cases

1. **Replace or Supplement On-Premises File Servers**: Azure Files can replace or supplement traditional file servers or NAS devices, allowing direct mounting from various operating systems.
2. **"Lift and Shift" Applications**: Facilitates moving applications to the cloud, either entirely or in a hybrid scenario where the application continues to run on-premises.
3. **Simplify Cloud Development**: Useful for shared application settings, diagnostic shares, and providing tools/utilities for Dev/Test/Debug scenarios.
4. **Containerization**: Supports persistent volumes for stateful containers, allowing containers to access a shared file system.

Azure Queues

Azure Queue Storage is a cloud-based service that stores large numbers of messages, facilitating communication between different parts of a distributed application. It's accessible via a specific URL format and hosted within a storage account. Each queue contains messages, which can be any format and up to 64 KB in size.

The service enables applications to scale instantly to meet demand, making it a flexible solution for varying workloads. It's particularly useful for asynchronous processing, load leveling, and batch processing. Messages can be stored and accessed globally via authenticated HTTP or HTTPS calls, and the service can be integrated with various Azure services and programming languages.

Azure Queue Storage not only simplifies inter-component communication but also provides robust mechanisms for handling large volumes of messages efficiently. Its architecture, benefits, and applications make it a vital tool for building scalable and distributed cloud-based solutions.

Azure Table Storage

Azure Table Storage is a cloud-based service that stores non-relational structured data, also referred to as structured NoSQL data. It offers a key/attribute store with a schemaless design, making it adaptable and cost-effective. This storage can handle vast amounts of data, including user information, address books, device details, and other metadata. It's particularly suitable for applications that don't need complex joins or foreign keys and can work with denormalized data for quick access.

The service allows for the storage of terabytes of structured data and is capable of serving web-scale applications. It enables quick querying using a clustered index and supports access through the OData protocol and LINQ queries. The structure of Table storage includes the storage account, tables, entities, properties, and URLs. Each entity, similar to a database row, can have up to 252 properties, along with three system properties for partition key, row key, and timestamp. This structure allows for efficient querying and atomic operations on data.

Azure Managed Disks

Azure Managed Disks are block-level storage volumes used with Azure Virtual Machines, acting like virtualized physical disks. They come in various types, including ultra disks, premium SSDs, standard SSDs, and standard HDDs, each catering to different needs and performance requirements.

Ultra disks are the highest-performing option, suitable for data-intensive workloads and offering up to 32-TiB per region. Premium SSDs provide high-performance for IO-intensive workloads, with a new version, Premium SSD v2, allowing more granular adjustments to performance without downtime. Standard SSDs are optimized for consistent performance at lower IOPS levels, ideal for web servers and lightly used applications. Standard HDDs offer reliable, low-cost support for latency-tolerant workloads.

Managed disks are designed for 99.999% availability, with three replicas of data for high durability. They allow scalable VM deployment, with up to 50,000 VM disks of a type in a subscription per region. There are three main disk roles in Azure: data disks for storing application data, OS disks containing the preinstalled OS, and temporary disks for short-term storage.

LRS vs ZRS

Azure Storage ensures data protection by storing multiple copies of data, and the redundancy options play a crucial role in determining the availability and durability of the data. Two primary redundancy options are Locally Redundant Storage (LRS) and Zone-Redundant Storage (ZRS).

LRS replicates data three times within a single data center in the primary region, providing 99.999999999% durability. It's the least expensive option but offers the least durability compared to other options. LRS is suitable for applications that can easily reconstruct data if lost or those restricted to replicating data within a specific region.

ZRS, on the other hand, replicates data across three Azure availability zones in the primary region, offering 99.9999999999% durability. Each availability zone is a separate physical location, ensuring that data remains accessible even if one zone becomes unavailable. ZRS is recommended for scenarios requiring high availability and is supported for various Azure Storage services, including Azure Blob storage, Azure Files, Azure Table storage, and Azure Queue storage.

Both LRS and ZRS handle write requests synchronously, ensuring that the write operation is successful only after the data is written to all replicas.

GRS vs GZRS

Azure Storage provides options for high durability by copying data to a secondary region, ensuring data remains intact even in the event of a complete regional outage. Two such options are Geo-redundant storage (GRS) and Geo-zone-redundant storage (GZRS).

GRS replicates data three times within a single location in the primary region and then asynchronously copies it to a single location in the secondary region, where it is again replicated three times. GZRS, on the other hand, synchronously replicates data across three availability zones in the primary region and then asynchronously copies it to the secondary region, following the same replication pattern as GRS.

Both GRS and GZRS offer 99.99999999999999% (16 9's) durability over a year. The main difference between them lies in how data is replicated in the primary region. GZRS is recommended for applications requiring maximum consistency, durability, availability, performance, and disaster recovery resilience.

Read-access configurations, RA-GRS and RA-GZRS, permit read access to the secondary region, ensuring data availability even if the primary region becomes unavailable. In the event of a primary region failure, a failover to the secondary region can be initiated, allowing read and write access.

Migrating Data in Azure

Moving Files in Azure

In Microsoft Azure, two essential tools for moving or migrating files are AzCopy and Storage Explorer.

AzCopy is a command-line utility used to copy blobs or files to or from a storage account. It is available in four versions for different operating systems, and authorization can be done using Azure AD or a Shared Access Signature (SAS) token. The method of authorization varies depending on the storage type, with options for Blob storage and File storage. AzCopy can be used as a standalone tool, in a script, or with Storage Explorer.

Storage Explorer is a standalone application that allows users to work with Azure Storage data on various platforms. It requires specific prerequisites depending on the operating system. After installation, users can sign into Azure or attach to an individual Azure Storage resource. Authentication options include using Azure AD for specific resource types or using the storage account's name and key, or a connection string with a SAS. Storage Explorer can also connect to individual storage resources using a SAS URI.

Both tools are designed to facilitate the transfer of data within Azure without delving into the technical details of the process, focusing on what they are and their capabilities.

Azure File Sync

Azure File Sync is a service that centralizes an organization's file shares in Azure Files, allowing for the flexibility, performance, and compatibility of a Windows file server. It can transform Windows Server into a quick cache of an Azure file share, supporting various protocols like SMB, NFS, and FTPS.

One of the key benefits of Azure File Sync is cloud tiering, where frequently accessed files are cached locally, and less frequently accessed files are stored in the cloud. This allows for seamless access while reducing costs by storing only a fraction of the data on-premises.

Azure File Sync also supports multi-site access and synchronization, making it suitable for distributed access scenarios. Changes made in one office are automatically synchronized across all other offices.

Additionally, Azure File Sync offers cloud-side backup, reducing on-premises backup costs. It leverages Azure Backup for centralized backups in the cloud, with native snapshot capabilities and automation for scheduling and managing backups. This integration ensures that any restored changes in the cloud are automatically downloaded to on-premises Windows Servers.

Azure Migrate

Azure Migrate serves as a centralized hub for assessing and migrating on-premises servers, infrastructure, applications, and data to Azure. It offers a variety of tools to facilitate different aspects of the migration process:

Azure Migrate: Discovery and Assessment: This tool helps discover and assess servers, including SQL and web apps, for migration to Azure. It works with VMware, Hyper-V, and physical servers.

Azure Migrate: Server Migration: This is used to migrate various types of virtual machines (VMs) and physical servers to Azure.

Data Migration Assistant: A stand-alone tool to assess SQL Server databases for migration to different Azure SQL services, identifying potential migration issues and benefits.

Azure Database Migration Service: This service enables the migration of on-premises databases to Azure VMs running SQL Server, Azure SQL Database, or SQL Managed Instances.

Movere: Acquired by Microsoft, this SaaS platform is used to assess servers and is available through specific Microsoft programs.

Azure App Service Migration Assistant: A stand-alone tool for assessing and migrating on-premises web apps to Azure.

Azure Data Box: A physical migration tool for transferring large amounts of offline data to Azure. Microsoft ships a physical storage device with a capacity of 80 TB, which is used to transfer data securely.

These tools collectively provide a comprehensive solution for various migration needs, from initial assessment to the actual transfer of data and applications.

Describe Azure Identity, Access, and Security

Azure Active Directory (Azure AD)

Azure Active Directory (Azure AD) is a cloud-based identity and access management service provided by Microsoft. It serves various users, including IT administrators, application developers, and subscribers to Microsoft services like Microsoft 365 and Azure.

Azure AD offers different licensing options, ranging from a free version with basic features to premium versions that include advanced administration, identity protection, and privileged identity management. The choice of license determines the availability of features such as single sign-on, multi-factor authentication, hybrid identity, and reporting and monitoring.

The service plays a vital role in managing user access to both internal and external resources, enforcing security measures like multi-factor authentication, and simplifying user experience through single signon capabilities. Azure AD is integral to many organizations for controlling access, enhancing security, and providing insights into usage patterns.

Azure Active Directory Domain Services (Azure AD DS)

Azure Active Directory Domain Services (Azure AD DS) is a managed version of traditional on-premises Active Directory, hosted in the cloud. It offers features like domain join, group policy, LDAP, and Kerberos and NTLM authentication. Azure AD DS integrates with existing Azure AD tenants, replicating identity information and making it compatible with both cloud-only and on-premises Azure AD tenants.

When deploying Azure AD DS, Azure creates a managed domain on a specified virtual network, spinning up two Windows server domain controllers. These controllers are fully managed by Azure, requiring no configuration or access by the user. The managed domain is configured for one-way synchronization, centralizing users, groups, and credentials. Resources should be created in Azure Active Directory to sync over to the managed domain service.

For organizations with an on-premises Active Directory, a hybrid configuration can be used. Users created in the on-premises directory are synchronized to Azure AD and then to the Azure AD DS managed domain. Applications, services, or virtual machines in Azure that connect to the Azure AD DS virtual network can use common Active Directory domain services features.

Azure AD DS offers several benefits, including easy deployment through a single wizard in the Azure portal, tight integration with Azure Active Directory, support for Kerberos and NTLM authentication, and high availability with multiple domain controllers. It's essential to understand that an Azure AD DS managed domain is a standalone domain, not an extension of an on-premises Active Directory domain, although one-way outbound forest trusts can be created if needed.

An Introduction to Azure AD B2B

Azure Active Directory B2B (Azure AD B2B), also known as business-to-business collaboration, is a service that enables organizations to share applications and services with guest users from other organizations. This service allows the host organization to retain control over its data while eliminating the need to manage external accounts, passwords, or synchronize accounts.

With Azure AD B2B, the partner organization uses its own identity management solution, and the host organization simply sends an email invitation to the guest user. If the guest user doesn't have an existing Microsoft account or Azure AD account, one is created upon redeeming the invitation. Authorization policies can then be applied to protect content, and guest users can be assigned to applications and groups like internal users.

Administrators can also delegate the management of guest users to application and group owners within the organization, reducing administrative overhead. Licensing for Azure AD B2B is based on the edition of Azure AD being used, with the ability to invite up to five guest users for each paid Azure AD license owned. The invited users' access to features depends on the edition of Azure AD used by the host organization.

An Introduction to Azure AD B2C

Azure Active Directory B2C (Azure AD B2C) is a business-to-consumer identity service designed to enable customers to sign on to applications using various account types, including social accounts like Facebook or LinkedIn, enterprise accounts, or local accounts. It serves as a white-label authentication solution, allowing businesses to customize the login experience to align with their brand.

The service leverages standard authentication protocols such as Open ID Connect, OAuth 2.0, and SAML, making it compatible with most modern applications and off-the-shelf software. Azure AD B2C provides a seamless login experience that integrates with the application's design, enhancing user convenience.

An existing Azure subscription is required to use Azure AD B2C, and there is a free tier available that supports up to 50,000 active users per month.

A Closer Look at SSPR, MFA, and Passwordless Authentication in Azure AD

Azure Active Directory (Azure AD) authentication offers several features to enhance the end-user experience, focusing on security and convenience. Self-service password reset (SSPR) enables users to change, reset, or unlock their accounts through a web browser, reducing the need for help desk intervention. Multi-factor authentication (MFA) adds an extra layer of security by requiring a second form of authentication, such as a phone call, SMS, or mobile app notification. Password protection in Azure AD includes a global banned password list and allows for custom password protection policies, ensuring the use of strong passwords. Passwordless authentication, such as Windows Hello or FIDO2 security keys, eliminates the need for passwords altogether, simplifying the sign-in process and reducing risks.

Azure AD Conditional Access

Azure Active Directory (Azure AD) Conditional Access is a security feature that enables control over who can access apps and data based on specified conditions. It functions as an if-then statement, where certain conditions must be met for access to be granted. The conditions, or signals, that Conditional Access policies rely on include User or Group Membership, Named location information, Device, Application, Real-time sign-in risk detection, Cloud apps or actions, and User risk. These signals allow for targeted policies, such as requiring multifactor authentication for specific user groups or applications, or blocking access from certain locations. Access controls within the policies determine the actions taken, either granting or blocking access, and may require extra verification. Conditional Access is available in paid editions of Azure AD, with specific features requiring different license levels, such as Azure AD Premium P1 or P2.

Azure Role-Based Access Control (RBAC)

Azure Role-Based Access Control (RBAC) is a system that manages permissions for Azure AD resource management. It's designed to allocate specific responsibilities to users, allowing for precise control over various administrative tasks within Azure AD. There are two types of roles in Azure RBAC: built-in roles and custom roles.

Built-in roles are preconfigured bundles of permissions designed for specific tasks. Some of the most common built-in roles include the Global Administrator role, which provides access to all administrative features; the User Administrator role, responsible for managing users and groups; and the Billing Administrator role, which handles purchases, subscriptions, and service health monitoring.

Custom roles offer more flexibility by allowing the creation of specific role definitions tailored to unique needs. These custom roles are created by selecting permissions from a preset list, the same ones used in built-in roles. Once defined, the custom role can be assigned to a user at either the organization-wide scope or an object scope, such as a single application. Custom roles require an Azure AD Premium P1 or P2 license.

In environments with Privileged Identity Management, role assignments can be either Eligible or Active. Eligible assignments require the user to perform an action, like multi-factor authentication, to use the role, while Active assignments grant the privileges without any additional action.

Understanding Zero Trust

The Zero-Trust Methodology is a security concept that emphasizes the need to trust no one and verify everything within an environment. It operates on the assumption that everything, even resources behind firewalls, is connected to an open and untrusted network. This approach means that traditional user passwords are no longer seen as sufficient for validation, and additional measures like multi-factor authentication (MFA) are implemented.

Three guiding principles underpin this model. The first, "Verify Explicitly," involves authenticating and authorizing based on all available data points. The second, "Least Privilege Access," limits user access with "just-in-time" and "just-enough" principles, along with risk-based adaptive policies. The third, "Assume Breach," focuses on segmenting access to resources, leveraging encryption, and using analytics for threat detection.

In addition to these principles, the Zero Trust model is supported by six foundational pillars that provide comprehensive security. These include the verification of identities with strong authentication, monitoring devices for health and compliance, discovering and managing all applications, classifying and encrypting data, assessing infrastructure for potential threats, and implementing network segmentation with real-time threat protection. Together, these principles and pillars work within the Zero Trust model to enforce organization security policies, strengthening the overall security posture of an organization.

The Defense in Depth Model

The defense in depth model is a multi-layered approach to security, designed to slow the advance of an attacker by providing several layers of protection. If one layer is breached, the next layer is there to prevent unauthorized access to data. The typical layers include physical security, identity and access, perimeter, network security, the compute layer, application layer, and data layer. Each layer has its specific role, from limiting physical access to the datacenter to encrypting business and customer data.

Another perspective on defense in depth is the Confidentiality, Integrity, and Availability (CIA) model. Confidentiality ensures that sensitive data remains private, typically through encryption. Integrity ensures that data or messages remain correct and haven't been tampered with. Availability ensures that data is accessible to those who need it. While all three aspects are vital, they may represent certain trade-offs in the security strategy.

What is Microsoft Defender for Cloud?

Microsoft Defender for Cloud is a comprehensive tool designed for security posture management and threat protection across cloud resources. It offers an array of Defender plans that provide defenses for various layers of the environment, including servers, storage, SQL, containers, and more. The tool helps in hardening resources, tracking security posture, protecting against cyber threats, and simplifying security management.

Defender for Cloud operates on three main principles: continuous assessment, secure, and defend. Continuous assessment helps in understanding the current security posture through a secure score, which provides a single metric to gauge the risk level. The secure principle focuses on hardening all connected resources and services, offering customized and prioritized tasks to improve the security posture. The defend principle is about detecting and resolving threats, with security alerts that can be sent to relevant personnel or integrated with other solutions.

The central feature of Defender for Cloud is the secure score, which continually assesses resources and aggregates findings into a single score to reflect the current security situation. When first opened, Defender for Cloud generates a secure score based on an assessment of connected resources and provides hardening recommendations to strengthen the security posture across various platforms.

Describe Features and Tools in Azure for Governance and Compliance

Azure Governance Methodologies

Azure Governance Methodologies encompass various tools and services to govern Microsoft Azure.
These include Azure Policy, Policy Initiatives, Role-Based Access Control (RBAC), Resource Locks, and Azure Blueprints. Azure Policy helps in creating, assigning, and managing policies to ensure compliance with corporate standards. Policy Initiatives work in conjunction with Azure policies to track compliance.
RBAC provides fine-grained access management, allowing specific permissions for different roles. Resource Locks prevent accidental deletion or modification of resources, and Azure Blueprints allow cloud architects to define repeatable sets of Azure resources adhering to organizational standards.

Privacy, Compliance, and Data Protection Standards

In this lecture, we explored various compliance terms, requirements, and services used to set and maintain privacy compliance and data protection in Microsoft Azure. Key topics included the Microsoft Privacy Statement, Trust Center, Service Trust Portal, Compliance Manager, Azure Government Services, and Azure China 21Vianet. We also discussed specific compliance offerings such as CJIS, CSA Star Certification, GDPR, HIPAA, and NIST. The lecture emphasized the importance of understanding how cloud providers can help comply with regulations and standards, and provided insights into Microsoft's robust compliance processes.

Azure Cost Management

Planning and Managing Costs

In the lecture, students are introduced to the various methods of purchasing Azure products and services, and the factors that influence these costs. Three main types of customers are identified: enterprise customers, who often have customized pricing and yearly agreements; web direct customers, who pay public prices monthly; and cloud solution providers (CSPs), who handle billing and payment directly. The lecture also explains how costs are affected by factors such as resource type, services, location, and usage rates. The concept of bandwidth and its cost implications are discussed, along with the geographical groupings of Azure regions known as zones. Tools like the Azure pricing calculator and the total cost of ownership (TCO) calculator are explained, providing ways to estimate and compare costs. Finally, various strategies and tools for minimizing and managing Azure costs are presented, including performing cost analysis, setting spending limits, utilizing Azure Reservations, and using the Azure cost management tool.

Monitoring and Reporting in Azure

In this lecture, students are introduced to the essential aspects of monitoring and reporting in Azure. The content begins with an explanation of tags, which are used to logically organize Azure resources by adding metadata. This helps in separating resources like production and development and is vital for billing or management. However, there are limitations to using tags, such as restrictions on the number of tags and characters allowed.

The lecture then delves into Azure Monitor, a service that collects, analyzes, and acts on telemetry from various environments. It's a comprehensive tool that offers features to analyze, respond, visualize, and integrate monitoring data. Azure Service Health is also discussed, providing personalized guidance and support for issues with Azure services. It consists of Azure Status, Service Health, and Resource Health, each serving a unique purpose in maintaining the health status of Azure.

Further, the lecture explores Application Insights for monitoring web applications, Azure Monitor for Containers for container workloads, and Azure Monitor for VMs for virtual machines. The importance of proactive responses to identified issues, such as alerts and auto-scale, is emphasized. The lecture concludes with insights into visualizations and integrations within Azure Monitor, highlighting the need to understand the differences and relationships between various Azure monitoring tools.